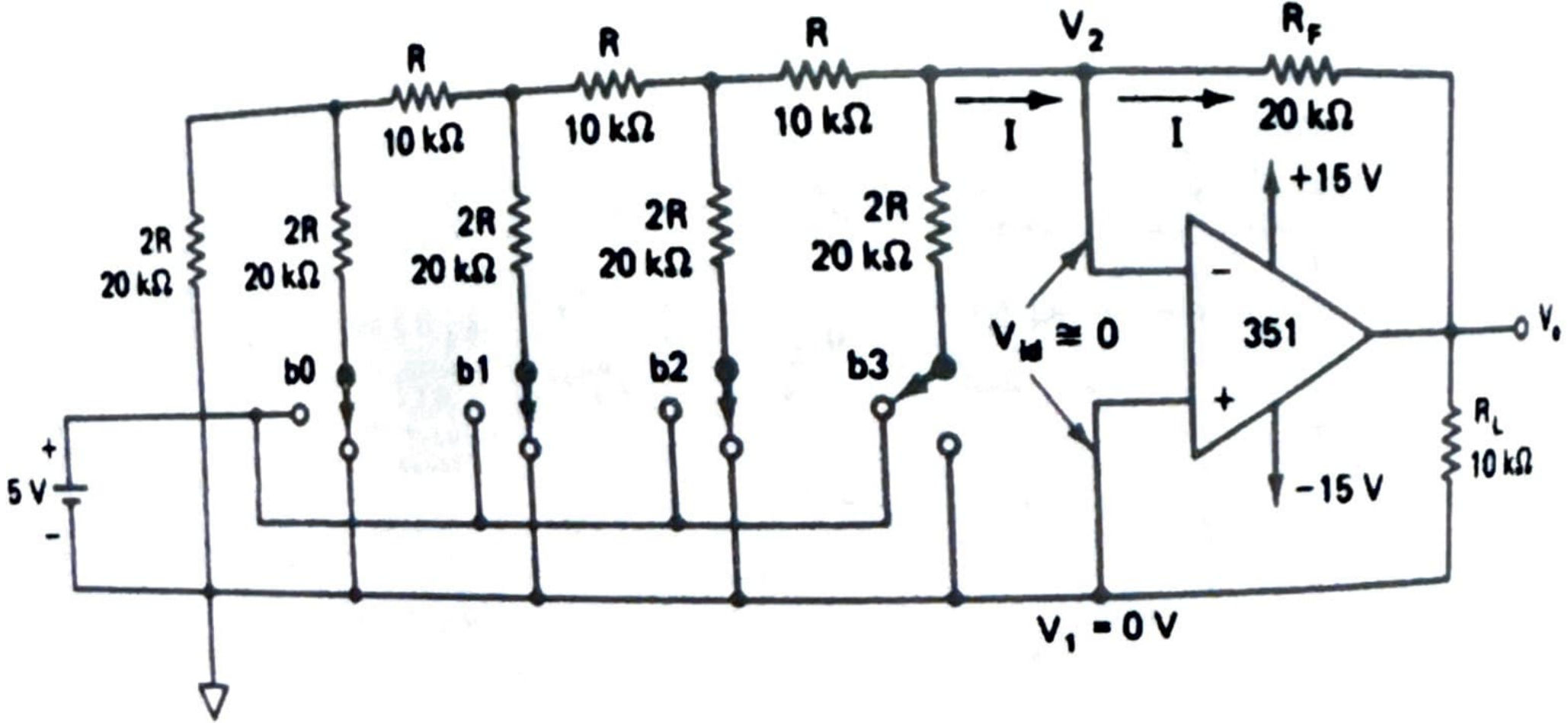
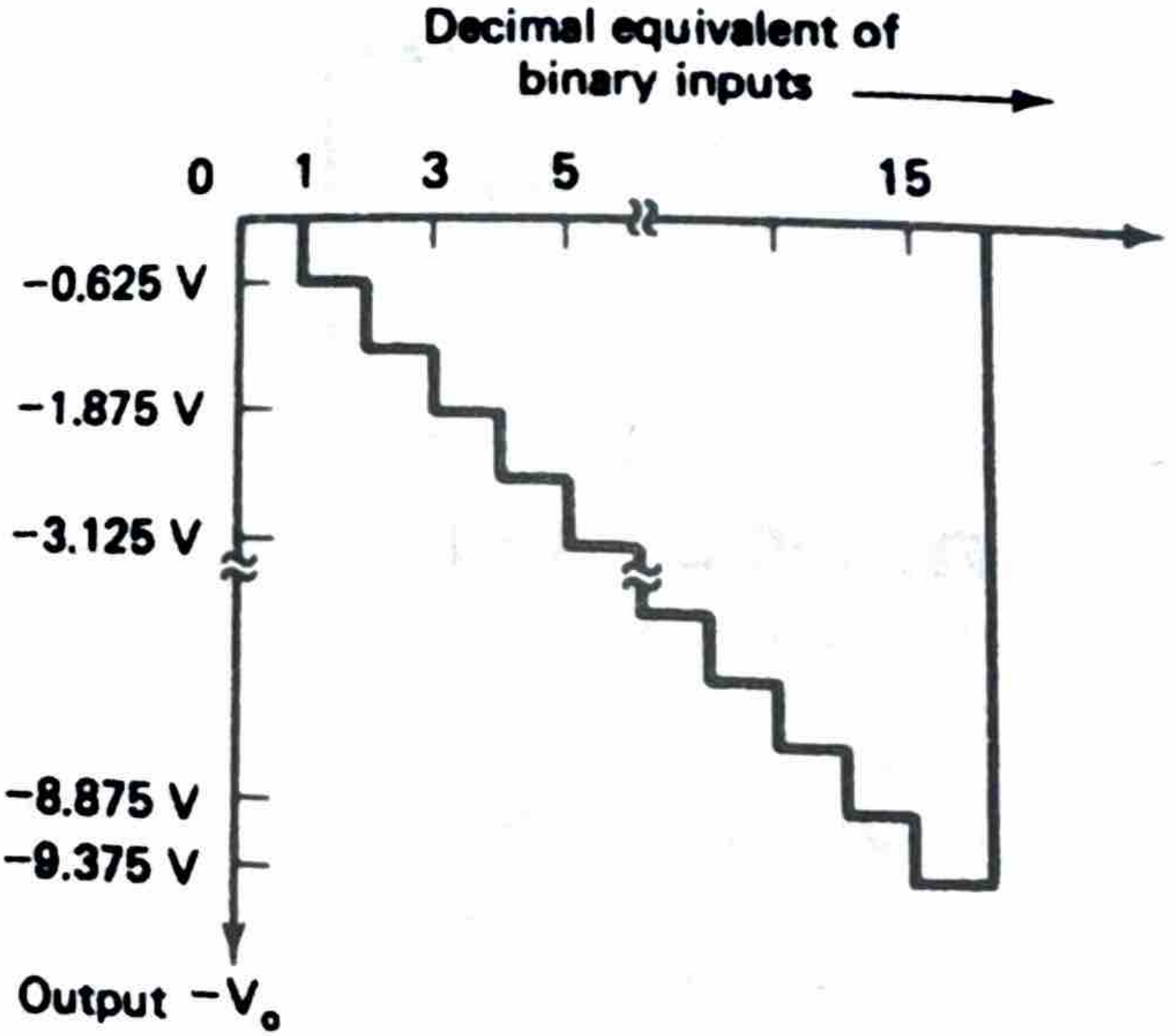

R
10 kΩ
R
10 kΩ
R
10 kΩ
V_2
R_F
20 kΩ
I
I
+15 V
2R
20 kΩ
2R
20 kΩ
2R
20 kΩ
2R
20 kΩ
2R
20 kΩ
b0
b1
b2
b3
$V_{id} \cong 0$
351
V_o
R_L
10 kΩ
+
5 V
-
-15 V
$V_1 = 0$ V

(a)


Decimal equivalent of
binary inputs
0
1
3
5
15
-0.625 V
-1.875 V
-3.125 V
-8.875 V
-9.375 V
Output $-V_o$

output voltage. When all the switches are closed, obviously the output will be maximum. The output voltage equation is given by

$$V_o = -R_F\left(\frac{\text{b0}}{R} + \frac{\text{b1}}{R/2} + \frac{\text{b2}}{R/4} + \frac{\text{b3}}{R/8}\right) \tag{8–6}$$

where each of the inputs b3, b2, b1, and b0 may either be high (+5 V) or low (0 V).

Figure 8–18(b) shows analog outputs versus possible combinations of inputs. The output is a negative-going staircase waveform with 15 steps of −0.5 V each. In practice, however, the steps may not all be the same size because of the variations in *logic high* voltage levels. Notice that the size of the steps depends on the value of R_F. Therefore, a desired step can be obtained by selecting a proper value of R_F, provided that the maximum output voltage does not exceed the saturation levels of an op-amp. For accurate operation of the D/A converter precision metal film resistors are recommended.

The problem with the D/A converter of Figure 8–18 is that it requires binary-weighted resistors, which may not be readily available, especially if the number of inputs is more than four. An attractive alternative is to use R and $2R$ resistors for the D/A converter since it requires only two sets of precision resistance values.